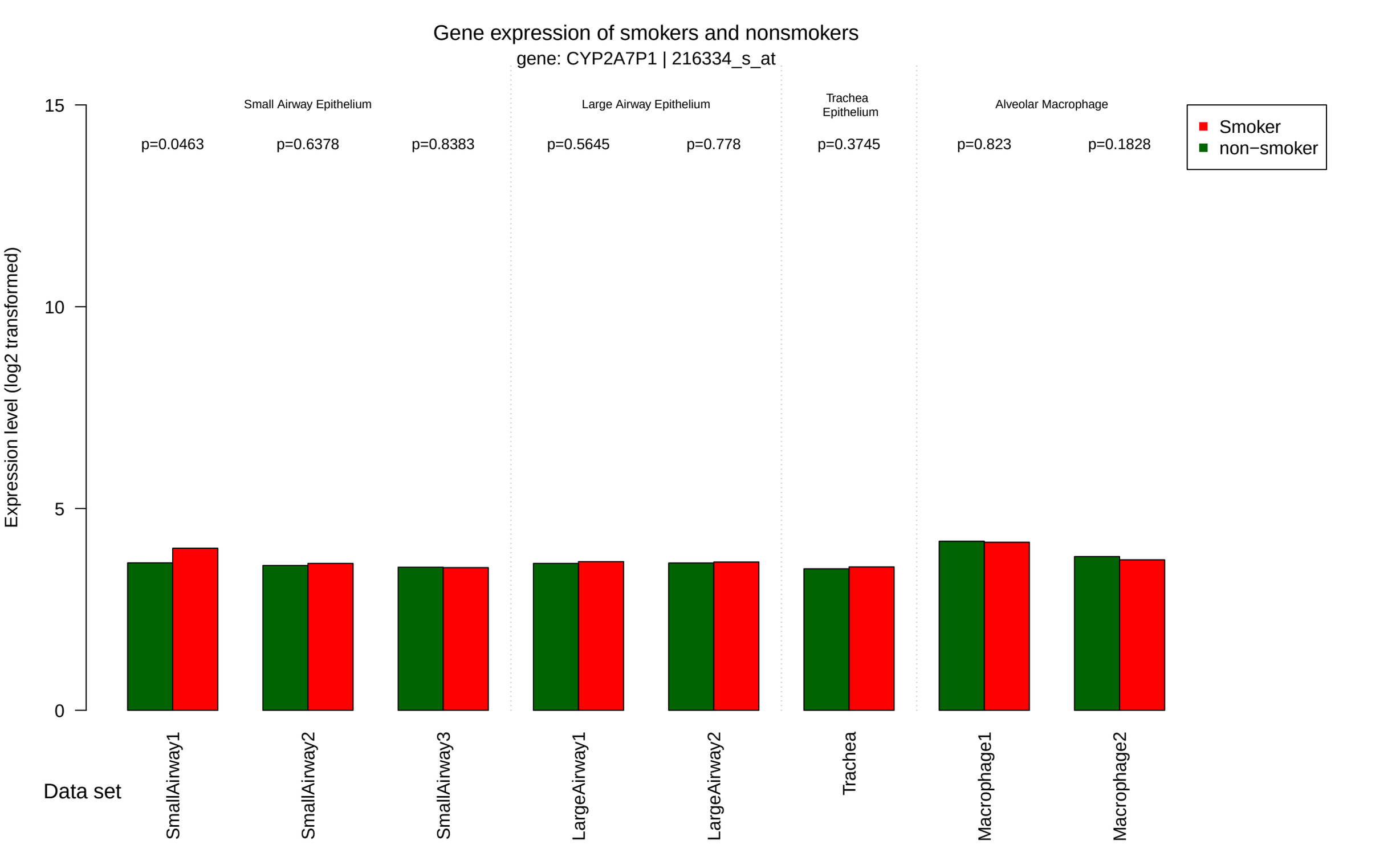
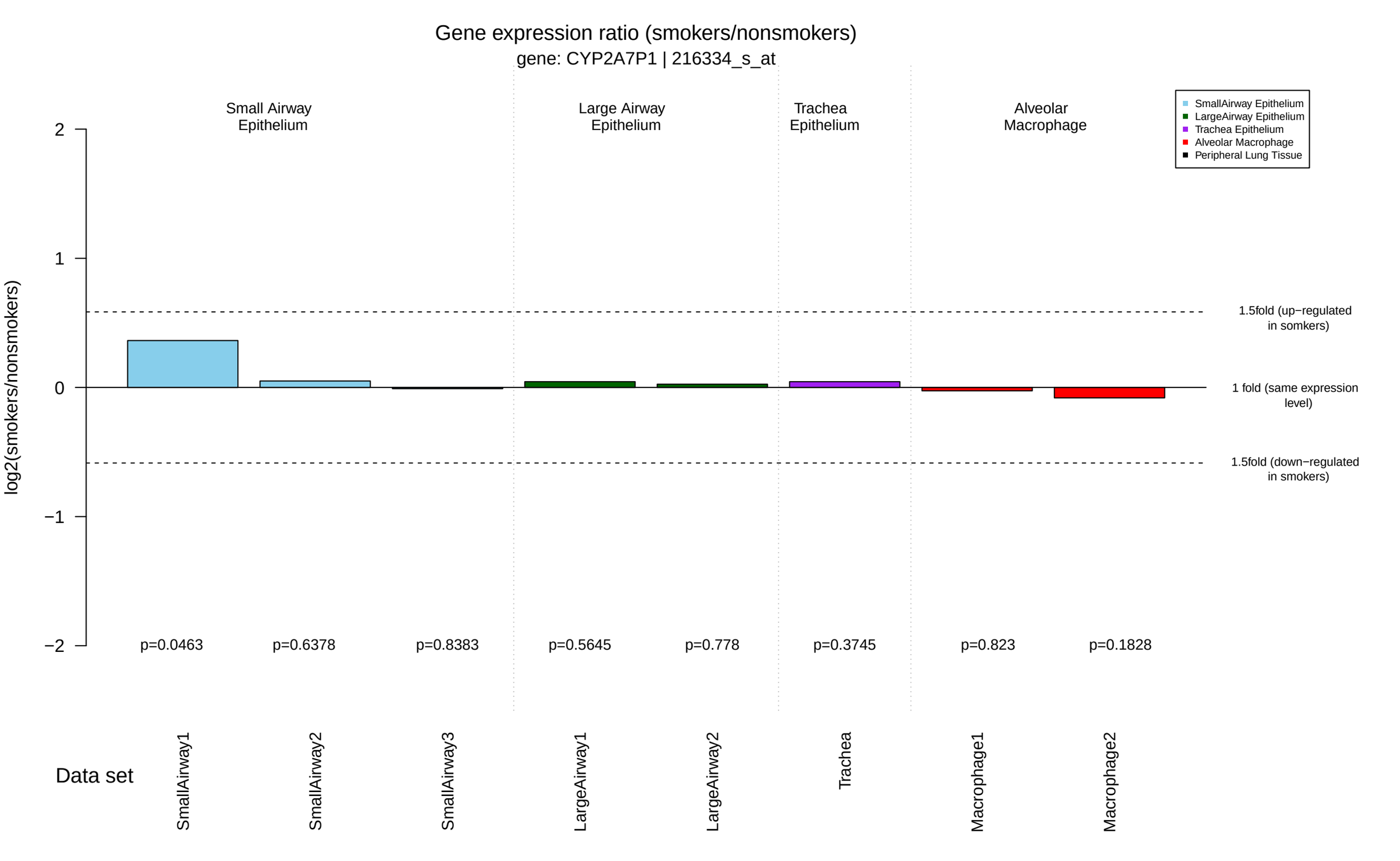
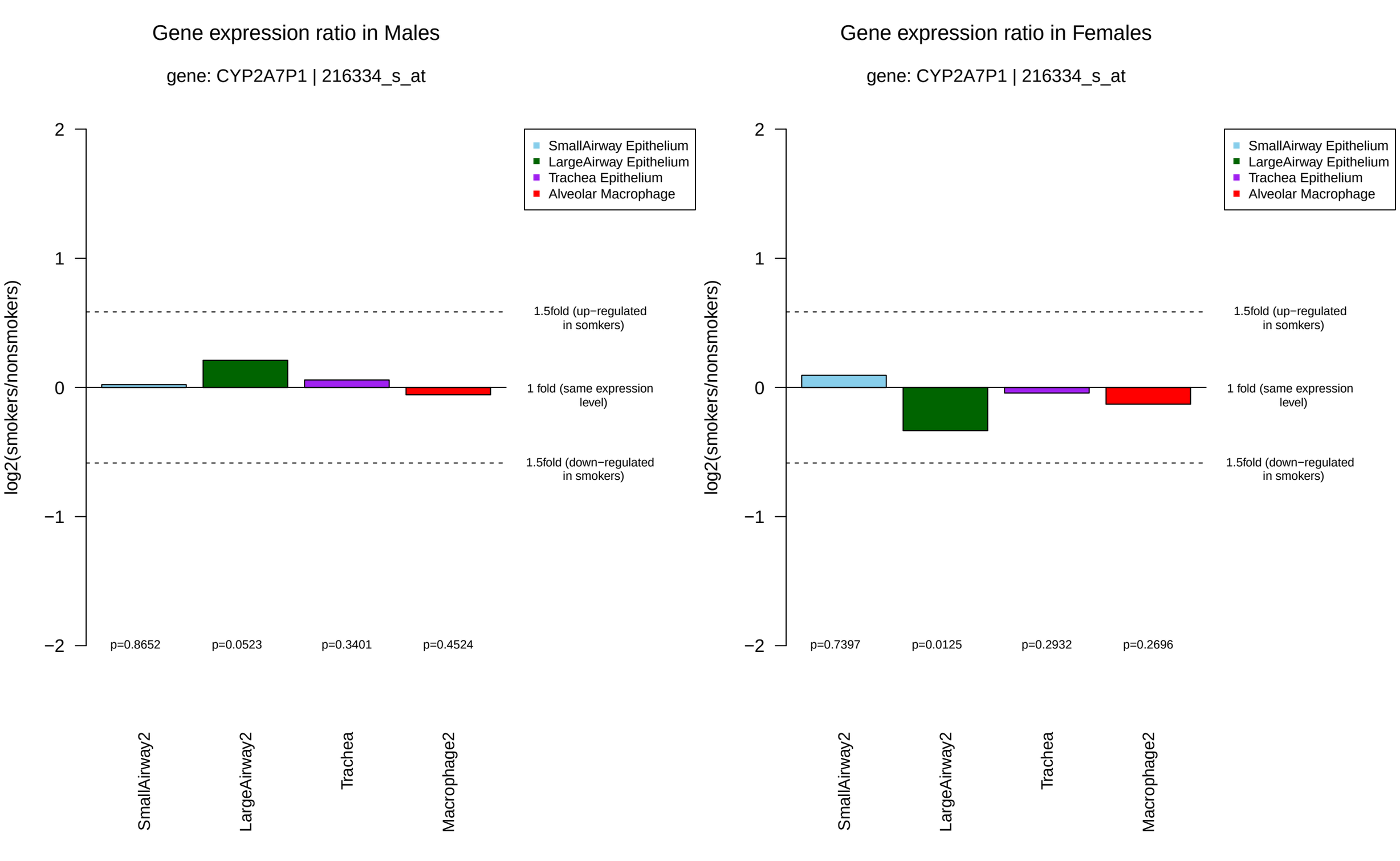
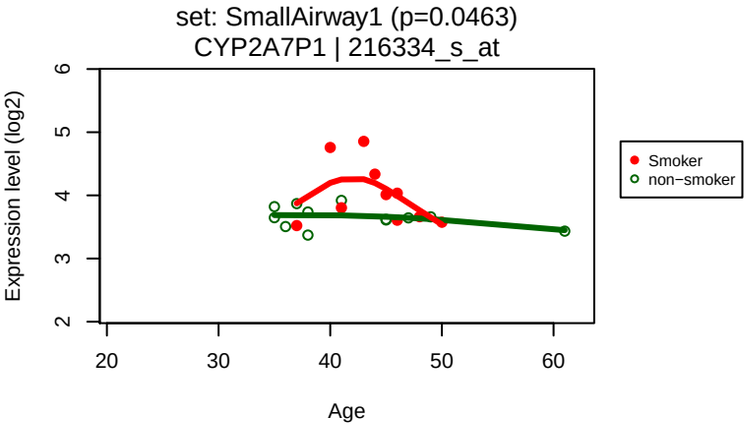
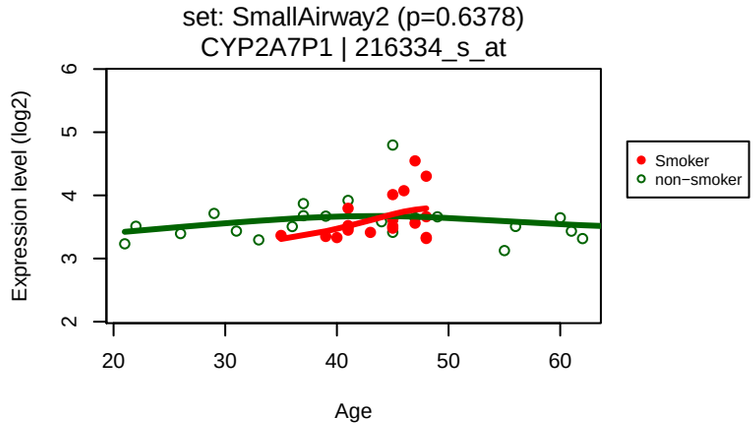
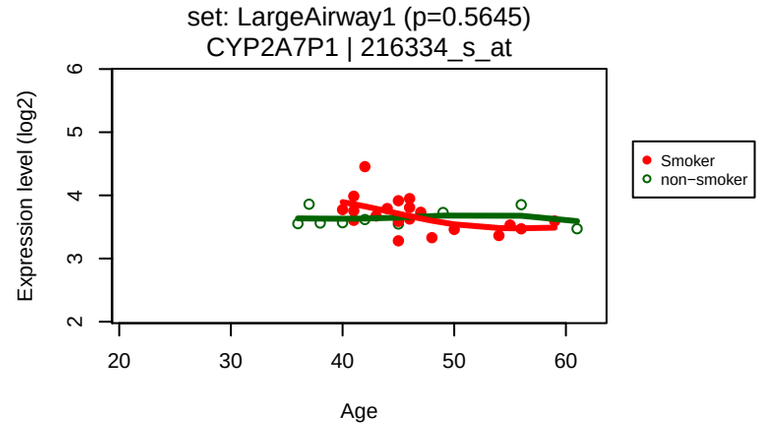

Gene expression of smokers and nonsmokers
gene: CYP2A7P1 | 216334_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0463
p=0.6378
p=0.8383
p=0.5645
p=0.778
p=0.3745
p=0.823
p=0.1828
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

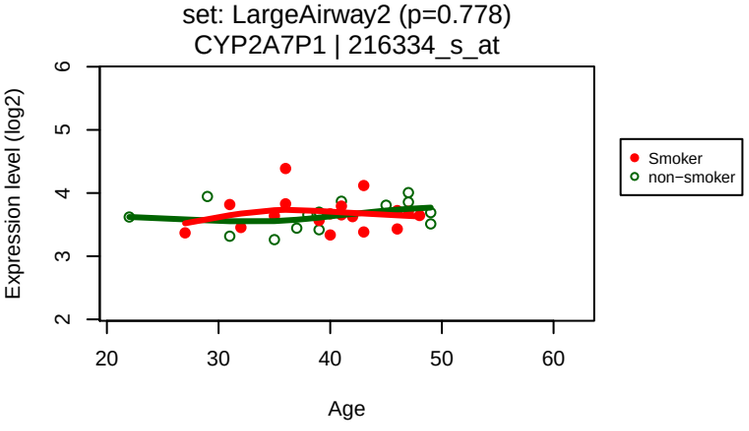

Gene expression ratio (smokers/nonsmokers)
gene: CYP2A7P1 | 216334_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0463
p=0.6378
p=0.8383
p=0.5645
p=0.778
p=0.3745
p=0.823
p=0.1828
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

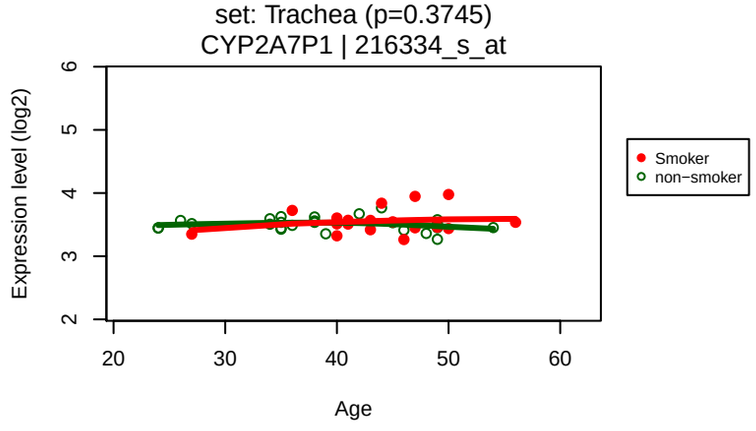

Gene expression ratio in Males
gene: CYP2A7P1 | 216334_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8652
p=0.0523
p=0.3401
p=0.4524
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2A7P1 | 216334_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7397
p=0.0125
p=0.2932
p=0.2696
SmallAirway2
LargeAirway2
Trachea
Macrophage2

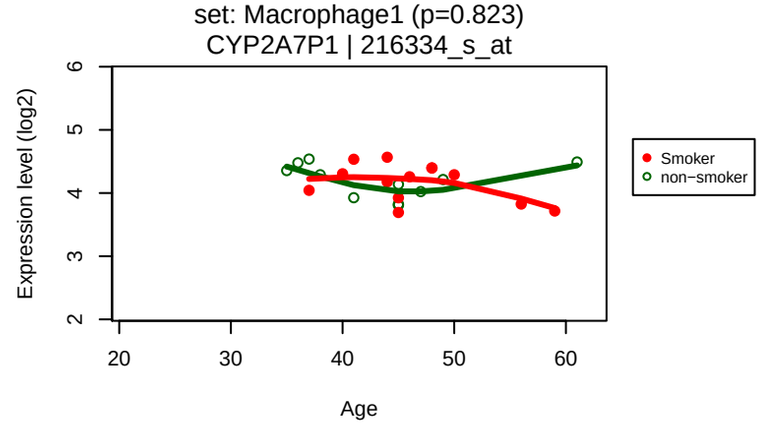

set: SmallAirway1 (p=0.0463)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker

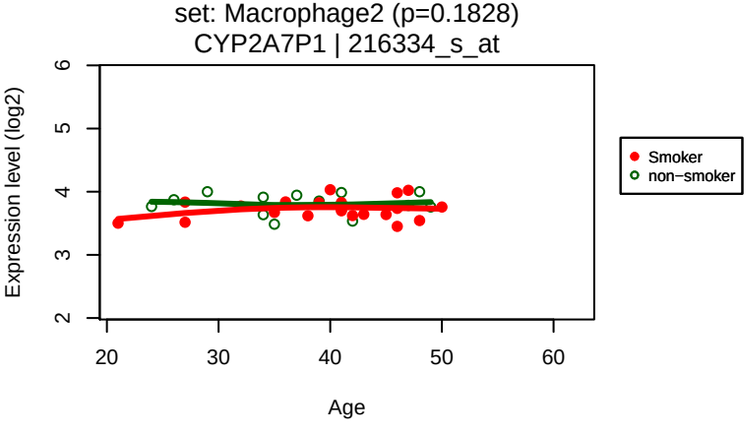

set: SmallAirway2 (p=0.6378)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker

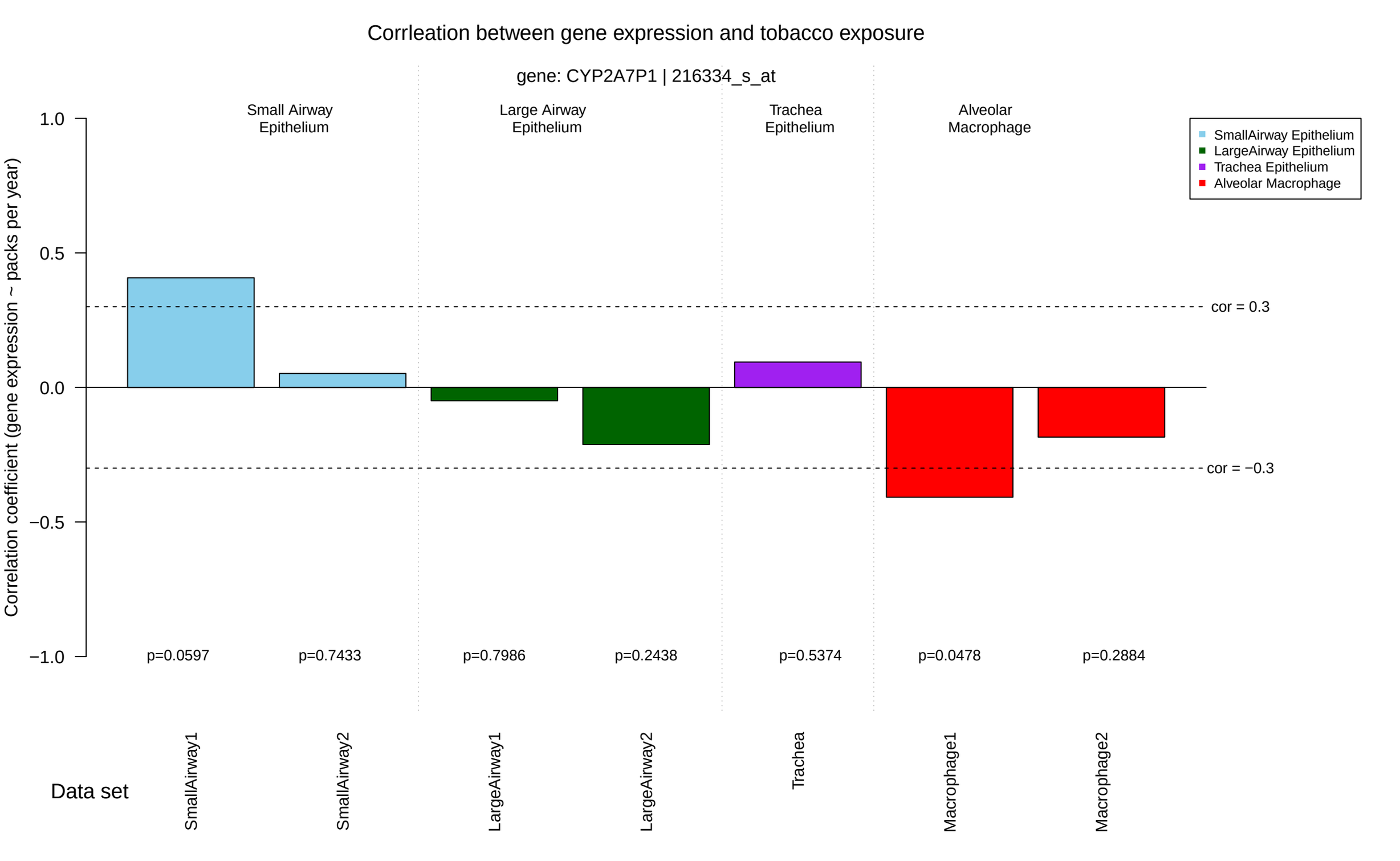

set: LargeAirway1 (p=0.5645)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.778)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3745)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.823)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1828)
CYP2A7P1 | 216334_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2A7P1 | 216334_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0597
p=0.7433
p=0.7986
p=0.2438
p=0.5374
p=0.0478
p=0.2884
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2